

1. **MS:** 4.3, 4.4, 4.8, 4.9, 5.2 (a, b, c, h), 6.1 (all), 6.2 (b, d, f, g, k, l), 6.3

2. **From Sydsaeter and Hammond**

6.3 Compute the first- and second-order derivatives of the following functions:

$$(a) y = 2x - 5 \quad (b) y = \frac{1}{3}x^9 \quad (c) y = 1 - \frac{1}{10}x^{10} \quad (d) y = 3x^7 + 8$$

$$(e) y = \frac{x-5}{10} \quad (f) y = x^5 - x^{-5} \quad (g) y = \frac{x^4}{4} + \frac{x^3}{3} + \frac{52}{2} \quad (h) y = \frac{1}{x} + \frac{1}{x^3}$$

6.5.4 Compute the following limits:

$$(a) \lim_{x \rightarrow 2} (x^2 + 3x - 5) \quad (d) \lim_{x \rightarrow 0} \frac{x^3 + 3x^2 - 2x}{x}$$

$$(b) \lim_{y \rightarrow -3} \frac{1}{y+8} \quad (e) \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

$$(c) \lim_{x \rightarrow 0} \frac{x^3 - 2x - 1}{x^5 - x^2 - 1} \quad (f) \lim_{x \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \quad (h \neq 0)$$

7.1.1 Find y' by implicit differentiation if $3x^2 + 2y = 5$. Check by solving for y and then differentiating.

7.1.2 For the equation $x^2y = 1$, find dy/dx and d^2y/dx^2 by implicit differentiation. Check by solving for y and then differentiating.

3. **From Gill**

5.8 Use L'Hospital's Rule to evaluate the following limits

$$(a) \lim_{x \rightarrow \infty} \frac{\ln(x)}{x} \quad (b) \lim_{x \rightarrow \infty} \frac{3^x}{x^3} \quad (c) \lim_{y \rightarrow \infty} \frac{3e^y}{y^3}$$

6.9 Obtain the first, second and third derivatives of the following functions:

$$(a) f(x) = 5x^4 + 3x^3 - 11x^2 + x - 7 \quad (b) h(z) = 111z^3 - 121z \quad (c) f(y) = \sqrt{y} + \frac{1}{y^{7/2}}$$

4. **Additional Questions:**

- (a) Suppose $\pi(Q) = QP(Q) - cQ$, where P is a differentiable function and c is a constant. Find an expression for $\frac{d\pi}{dQ}$.
- (b) Suppose $\pi(L) = PF(L) - wL$, where F is a differentiable function and P and w are constants. Find an expression for $\frac{d\pi}{dL}$.